Sampling and confidence intervals CHEAT SHEET

SAMPLING

population: every element · sample: those measured

frame: the list the sample is drawn from

left off the frame → the sample can't speak for them

parameter μ, σ, p (fixed, unknown)

statistic $\bar{x}$, s, $\bar{p}$ (varies from sample to sample)

simple random sample: every sample of n equally likely

$C(8, 3) = 56$ samples · $C(12, 4) = 495$

RAND beside each row, sort, take the first n

infinite (a process): same population, independent

point estimates: $\bar{x} \to \mu$, $s \to \sigma$, $\bar{p} \to p$

sampling error $= \bar{x} - \mu$ (this sample: $81.75 - 80 = 1.75$)

OTHER DESIGNS AND SURVEY ERRORS

stratified: alike inside strata; $\bar{x}_{st} = \Sigma \frac{N_h}{N} \bar{x}_h$

$0.6 \times 310 + 0.3 \times 1450 + 0.1 \times 4100 = \$1{,}031$

(the plain mean of the three, \$1,953, misleads)

cluster: whole clusters (cheaper, but wider)

systematic: every k-th, $k = N/n$, random start

5,000 orders, $n = 50$: $k = 5000/50 = 100$

convenience, judgement: no probability → no margin

sampling error shrinks as n grows

nonsampling error (frame, nonresponse, wording, measurement) does not. Size cannot fix bias.

SAMPLING DISTRIBUTIONS

$E(\bar{x}) = \mu$ $\quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ (the standard error)

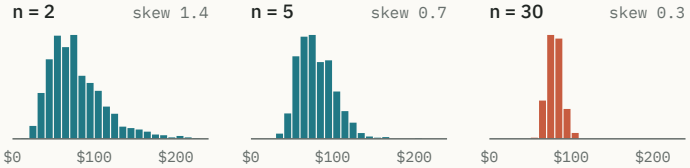
orders: $\sigma = \$50$, $n = 30$: $\frac{50}{\sqrt{30}} \approx 9.13$; simulated: 9.05

$4 \times n \to$ ½ SE: $n = 120$: $\frac{50}{\sqrt{120}} \approx 4.56$

finite: $\times \sqrt{(N - n)/(N - 1)}$ when $n/N > 5\ \%$

100 of 400: $\times 0.867$ · 30 of 5,000: $\times 0.997$ (ignore)

CLT: $\bar{x} \approx$ normal for large n ($n \geq 30$; more if skewed)



1,000 means each · skew 1.9 → 1.4, 0.7, 0.3

$E(\bar{p}) = p$ $\quad \sigma_{\bar{p}} = \sqrt{p(1 - p)/n}$; np, $n(1 - p) \geq 5$

express 30 %, $n = 100$: $\sqrt{0.3 \times 0.7/100} = 0.0458$

$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$: fares \$24, \$8, $n = 64$: $P(\bar{x} > 25.5) = 0.0668$

GOOD ESTIMATORS

unbiased: $E(\hat{\theta}) = \theta$ ($\bar{x}$, $\bar{p}$, s^2 with $n - 1$)

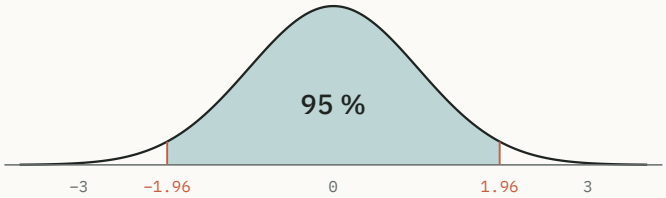
efficient: the smaller standard error

normal data, $n = 25$: mean 0.8 g vs median ≈ 1.0 g

consistent: SE → 0 as n grows

1.26, 0.63, 0.32 at $n = 10, 40, 160$

THE NORMAL CURVE AND Z



$z = \frac{x - \mu}{\sigma}$ $\quad$ 68 % · 95 % · 99.7 % within 1, 2, 3 σ

$\mu = 50$, $\sigma = 4$: $x = 56 \to z = 1.5 \to P(z \leq 1.5) = 0.9332$

NORM.DIST(56, 50, 4, TRUE) = 0.9332

NORM.S.DIST(1.5, TRUE) = 0.9332

middle 95 %: ± 1.96 = NORM.S.INV(0.975)

PITFALLS

a big convenient sample is still biased

the latest 300 invoices: 4.3 % (the random 300: 10 %)

$n/N > 5\ \%$: use the correction (100 of 400: SE $\times 0.867$)

σ unknown, small n: t, not z (z covers ≈ 92 % at $n = 10$)

not '95 % chance μ is in this interval':

the method covers μ in 95 % of samples

sample sizes: always round up (138.3 → 139)

Sampling and confidence intervals CHEAT SHEET

THE MEAN, SIGMA KNOWN

σ known: $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$ $z = 1.645 \cdot 1.96 \cdot 2.576$

for 90 % · 95 % · 99 % confidence

pick time: $12.4 \pm 1.96 \times \frac{1.5}{6} = 12.4 \pm 0.49$ min

CONFIDENCE.NORM(0.05, 1.5, 36) = 0.49

THE MEAN, SIGMA UNKNOWN

σ unknown: $\bar{x} \pm t \frac{s}{\sqrt{n}}$ $df = n - 1$

loans: $12 \pm 2.262 \times \frac{2.58}{\sqrt{10}} = 12 \pm 1.85$ days

z with s at n = 10 covers only ≈ 92 %

T.INV.2T(0.05, df) · CONFIDENCE.T(0.05, s, n)

small n: the population roughly normal (plot it)

df	5	9	15	24	29	63	299	∞
t	2.571	2.262	2.131	2.064	2.045	1.998	1.968	1.960

THE CASE: AN INVOICE AUDIT

Fernmoor Health (fictional): 4,000 invoices

a random sample of 300

error rate 30/300 = 10 % ± 3.3 (with × 0.962)

→ 6.7 % to 13.3 % of invoices in error

total = N $\bar{x}$ = 4000 × $9.66 ≈ $38,600 ± $16,800

the latest 300 alone said 4.3 %: a convenient sample misleads, and no margin shows it

INTERVAL FOR A PROPORTION

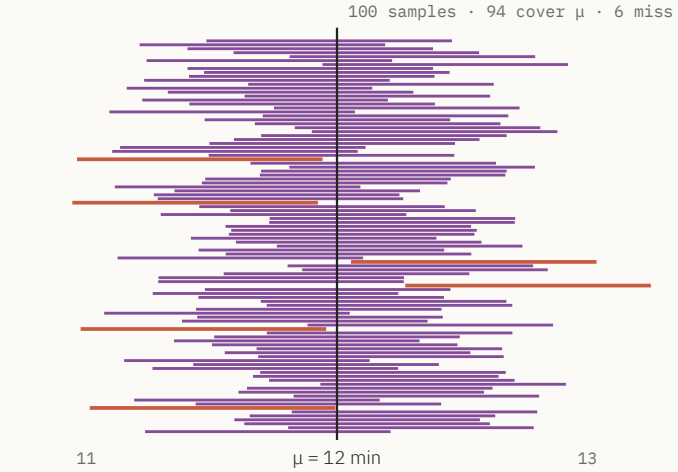
$\bar{p} \pm z \sqrt{\bar{p}(1-\bar{p})/n}$ check $n\bar{p}$, $n(1-\bar{p}) \geq 5$

420 of 1,000: 0.42 ± 0.0306 → 38.9 % to 45.1 %

$N = 10^7, 10^5, 10^4$ → ± 3.06, 3.04, 2.90 points:

the population's size barely matters

WHAT 95 % CONFIDENCE MEANS



each bar: $\bar{x} \pm 0.49$ from a new sample of 36

95 %: the method covers μ in 95 of 100 samples (here 94)

READING AN INTERVAL

report: estimate ± margin, level, n, how drawn

width grows with $\frac{z}{\sqrt{n}}$: 99 % vs 95 %: $\times \frac{2.576}{1.96} \approx 1.31$

4 × n: half as wide ($\sqrt{4} = 2$)

ask: how was it drawn? what is the margin?

and what bias could no margin cover?

SAMPLE SIZE

mean: $n = (\frac{z\sigma}{E})^2$ $(1.96 \times 1.5/0.25)^2 = 138.3 \rightarrow 139$

planning σ: earlier study, pilot, or range/4

½ E → 4 × n: E = 0.125 → 554

proportion: $n = \frac{z^2\, p^*(1-p^*)}{E^2}$ $p^* = 0.5$ if unknown

± 3 points, 95 %: 1,068 · ± 2 points, 99 %: 4,148

finite: $n' = \frac{n}{1 + (n-1)/N}$ N = 2,000: 1068 → 697

always round up

SPREADSHEET FUNCTIONS

RAND(): a number per row; sort, take the first n

COMBIN(8, 3) = 56 samples · COUNT, COUNTA: n

COUNTIF, COUNTIFS: how many say yes (30 of 300)

AVERAGE: $\bar{x}$ · AVERAGEIFS: one stratum's mean

SUM, SUMPRODUCT: $\Sigma\, N_h/N \times \bar{x}_h$ = $1,031

STDEV.S, VAR.S (n − 1): loans s = 2.58, s^2 = 6.67

NORM.DIST(56, 50, 4, TRUE) = 0.9332

NORM.S.DIST(1.5, TRUE) = 0.9332

NORM.S.INV(0.975) = 1.96

CONFIDENCE.NORM(0.05, 1.5, 36) = 0.49

T.INV.2T(0.05, 9) = 2.262

T.DIST(1.96, 9, TRUE) = 0.959 (2 × 0.959 − 1 ≈ 92 %)

CONFIDENCE.T(0.05, 2.58, 10) = 1.85

SQRT · ABS · IF · ROUNDUP(138.3, 0) = 139